

Lecture 6

Let G be a group & let $H \leq G$. For $a, b \in G$

$a \equiv_r b \pmod{H}$ if and only if $ab^{-1} \in H$.

$\equiv_r$ is an equivalence relation on G .

Proof: $a \equiv_r a \pmod{H}$ as $a \cdot a^{-1} = 1 \in H$.

so $\equiv_r$ is reflexive.

$a \equiv_r b \pmod{H} \Rightarrow ab^{-1} \in H \Rightarrow (ab^{-1})^{-1} = (b^{-1})^{-1}a^{-1} = ba^{-1} \in H$

$$\Rightarrow b \equiv_r a \pmod{H}$$

so $\equiv_r$ is symmetric.

$a \equiv_r b \pmod{H}$ & $b \equiv_r c \pmod{H}$

$$\Rightarrow ab^{-1} \in H \quad bc^{-1} \in H \Rightarrow (ab^{-1})(bc^{-1}) \in H \Rightarrow ac^{-1} \in H$$
$$\Rightarrow a \equiv_r c \pmod{H}$$

Hence $\equiv_r$ is an equivalence relation.

Similarly we can define another equivalence relation on G

by saying $a \equiv_l b \pmod{H}$ if and only if $a^{-1}b \in H$.

The equivalence classes partition the group G .

So what does the equivalence classes look like?

Let $a \in G$. The equivalence class of a under $\equiv_r$ is defined by

$$\begin{aligned} [a]_r &= \{ b \in G \mid a \equiv_r b \pmod{H} \} \\ &= \{ b \in G \mid ab^{-1} \in H \} \\ &= \{ b \in G \mid ab^{-1} = h \text{ for } h \in H \} \end{aligned}$$

$$T - r^2 = 2 \cdot 1^2 = 2$$

21
13
23

3
2
1

Theorem: Let G be a group. $H \leq G$ and let $a, b \in G$.

Then the following hold.

- i) $Ha = H$ if and only if $a \in H$
- ii) $aH = H$ if and only if $a \in H$
- iii) $Ha = Hb$ if and only if $ab^{-1} \in H$
- iv) $aH = bH$ if and only if $a^{-1}b \in H$

Proof: i) $Ha = H = H \cdot 1 = \{h \cdot 1 \mid h \in H\}$

So $Ha = H$ if and only if $a \equiv_r 1 \pmod{H}$

$$\Rightarrow a^{-1} \in H$$

$$\Rightarrow a \in H.$$

ii) $aH = H = 1 \cdot H = \{1 \cdot h \mid h \in H\}$

So $aH = H$ if and only if $[a]_H = [1]_H$

$$\Rightarrow a \equiv_H 1 \pmod{H}$$

$$\Rightarrow a^{-1} \cdot 1 \in H$$

$$\Rightarrow a^{-1} \in H$$

$$\Rightarrow a \in H.$$

iii) $Ha = Hb \Leftrightarrow [a]_H = [b]_H$

$$\Leftrightarrow a \equiv_H b \pmod{H} \Leftrightarrow ab^{-1} \in H$$

iv) $aH = bH$ if and only if $[a]_H = [b]_H$

$$\Rightarrow a \equiv_H b \pmod{H}$$

$$\Rightarrow a^{-1}b \in H.$$

$$= \{b \in G \mid a = hb \text{ for some } h \in H\}$$

$$= \{b \in G \mid b = a^{-1}ha \text{ for some } h \in H\}$$

$$= \{ha \mid h \in H\} = aH.$$

This is called the right coset of a under H .

Similarly the equivalence class of a under $\equiv_L$ is defined by

$$[a]_L = \{b \in G \mid a \equiv_L b \pmod{H}\}$$

$$= \{b \in G \mid a^{-1}b \in H\}$$

$$= \{b \in G \mid a^{-1}b = h \text{ for some } h \in H\}$$

$$= \{b \in G \mid ab = ah \text{ for some } h \in H\}$$

$$= aH.$$

This is called left cosets of a under H .

Remark: For $a, b \in G$,

i) Either $aH \cap bH = \emptyset$ or $aH = bH$

ii) Either $Ha \cap Hb = \emptyset$ or $Ha = Hb$.

So the left cosets of H ~~are~~ form a partition of G .

Similarly the right cosets of H form a partition of G .

Example Remark: If a group G is abelian and $H \leq G$,
then we have $aH = Ha$ for all $a \in G$.

$$aH = \{ah \mid h \in H\}. \quad \text{Since } G \text{ is abelian,}$$

$$Ha = \{ha \mid h \in H\}. \quad ah = ha \quad \forall a, h \in G.$$

$$\text{So } aH = Ha \quad \forall a \in G.$$

Example: ① Let $(\mathbb{Z}, +)$ be a group. $m \in \mathbb{N}$.

$$\text{Let } H = \{mk \mid k \in \mathbb{Z}\}. \quad \text{Then } H \leq G.$$

$$0 \in \mathbb{Z}. \quad \text{Left coset of } 0 \text{ is } 0+H = \{0+mk \mid k \in \mathbb{Z}\} \\ = \{mk \mid k \in \mathbb{Z}\} = H.$$

$$1 \in \mathbb{Z}. \quad \text{Left coset of } 1 \text{ is } 1+H = \{1+mk \mid k \in \mathbb{Z}\}. \\ \text{Right coset of } 1 \text{ is } H+1 = \{mk+1 \mid k \in \mathbb{Z}\}.$$

② Take $G = S_3$ (Permutation group of 3 elements).

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

$$H = \{e, (12)\}.$$

H is a subgroup of S_3 as $(12)^{-1} = (12)$.

$$(13) \in S_3. \quad (13)H = \{(13) \cdot e, (13)(12)\} \\ = \{(13), (123)\}$$

$$H(13) = \{e(13), (12)(13)\}$$

$$= \{(13), (132)\}$$

Notice $(13)H \neq H(13)$

In general, left coset and right coset of an element need not be the same set.

Normal Subgroup: A subgroup H of a group G is called a normal subgroup of G if it satisfies

$$aH = Ha \quad \forall a \in G. \text{ and we write } H \trianglelefteq G.$$

Example: ① If a group G is abelian, then every subgroup of G is a normal subgroup.

② Consider $G = S_3$.

$$H = \{e, (132), (123)\}$$

Check H is a subgroup of G .

(Hint: Since H is finite, to prove H is a subgroup, we need to prove H is closed).

Check H is also a normal subgroup of G .